

Literature Review and Dataset Selection

Fraud Detection and AML Alert Triage

Client: STADIOEquities

This review identifies three publications that address a problem statement similar to this project, which is the use of machine learning to distinguish genuine financial crime cases from false positives in transaction-monitoring and fraud-detection systems. At least one of the publications uses a publicly available dataset that this project will also use. That publication is Publication 1. The final table describes the dataset that will be used for modelling in Part B.

Publication 1

Title	Enhancing credit card fraud detection using transactional and deep learning models with class imbalance mitigation
Harvard reference	Albalawi, T. and Dardouri, S. (2025) 'Enhancing credit card fraud detection using traditional and deep learning models with class imbalance mitigation', <i>Frontiers in Artificial Intelligence</i> , 8, 1643292. doi:10.3389/frai.2025.1643292.
Link	https://www.frontiersin.org/articles/10.3389/frai.2025.1643292/full
Modelling	The study trained three traditional classifiers, which were logistic regression, a decision tree, and a random forest. It also trained a deep learning neural network model. The deep learning model used focal loss so that training concentrated on the fraudulent cases that are hard to classify. The authors applied hyperparameter tuning to improve each model. The random forest produced the best overall result, with accuracy of 99.95 percent, an F1 score of 0.826, and a ROC-AUC of 0.976. The deep learning model produced the highest precision, which means it produced the fewer false positives.
Preprocessing techniques	The authors scaled and standardised the numeric input feature. They applied SMOTE (the Synthetic Minority Over-sampling Technique) to the training data in order to correct the severe class imbalance. They then split the data into training and test set for evaluation.
Datasets used	The study used two publicly available datasets. The first was the Kaggle Credit Card Fraud Detection dataset from the ULB Machine Learning Group. The second was the PaySim synthetic mobile-money

	dataset. Using both datasets allowed the authors to test whether their approach generalized across more than one domain.
Key takeaway	The first takeaway is that accuracy is a misleading metric on imbalanced fraud data, so this project will report precision, recall, F1, and ROC-AUC instead. The second takeaway is that SMOTE improves detection of the minority fraud class, so it is a sensible imbalance strategy for this project to adopt. The third takeaway is the technique which focus on the hardest and rarest cases, such as focal loss or class weighting, reducing false positive, which is the central goal of the project.

Publication 2

Title	Anti-Money Laundering Alert Optimisation Using Machine Learning with Graphs
Harvard reference	Naser Eddin, A., Bono, J., Aparício, D., Polido, D., Ascensão, J.T., Bizarro, P. and Ribeiro, P. (2022) 'Anti-Money Laundering Alert Optimization Using Machine Learning with Graphs', arXiv preprint arXiv:2112.07508. doi:10.48550/arXiv.2112.07508
Link	https://arxiv.org/abs/2112.07508 / https://arxiv.org/html/2112.07508v3
Modelling	The authors built a supervised machine learning triage model that works alongside the existing rule-based AML system. The model predicts the risk of each alert so that false positives can be suppressed and true positives can be prioritised. It uses gradient-boosted tree models trained on two kinds of features. The first kind is entity-centric engineered features, and the second kind is graph-based features that describe the relationship between entities. The authors reported a reduction in false positives of about 80 percent while retaining most of the true positives.
Preprocessing techniques	The authors aggregated transactional-level records into daily account-level features, including the total amounts sent and received, the counterparties, the timestamps, and the direction of each transaction. They extrapolated account level labels from the transaction level labels. They constructed a dynamic graph using timed windows and kept a reduced memory of legitimate entities to save time and space. They also accounted for delay in arrival of labels
Datasets used	The study used a real-world banking dataset that is proprietary and not publicly available. The data was labelled at the transaction level according to whether each transaction formed part of a Suspicious Activity Report.

Key takeaway	The first takeaway is that treating the task as a triage layer on top of the existing rule, which predicts alert risk rather than replacing the rules, is the same approach this project takes, the second takeaway is that ranking alerts by predicted risk is a more practical output for investigators than a simple accept or reject decision. The third takeaway is that aggregating raw transaction into account-level features is a useful preprocessing pattern that this project can reuse.
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Publication 3

Title	Machine Learning in Transaction Monitoring: The Prospect of Xai
Harvard reference	Gerlings, J. and Constantiou, I. (2023) 'Machine Learning in Transaction Monitoring: The Prospect of xAI', in Bui, T.X. (ed.) Proceedings of the 56th Hawaii International Conference on System Sciences (HICSS). Honolulu: HICSS, pp. 3474–3483. Also available as arXiv:2210.07648.
Link	https://arxiv.org/abs/2210.07648 https://arxiv.org/pdf/2210.07648
Modelling	This publication is a qualitative empirical study rather than a paper that trains a single classifier. It examines how machine learning affects the transaction-monitoring process when it either augments or fully automates the investigators work. It then examines the requirements for explainable AI (xAI) around such models. The authors find that the explainability requirements change depending on which party is held liable in the process.
Preprocessing techniques	This criteria does not apply in the usual modelling sense because the study focuses on the process and on explainability requirements rather than a data pipeline. The publication is included because it is directly relevant to how an alert-triage model should be presented to, and trusted by, compliance investigators
Dataset used	The study does not use a public modelling dataset. It is based on empirical research carried out within a financial institution transaction monitoring context.
Key takeaways	The first takeaway is that the output of a triage model must be explainable to investigators and auditors, because clear explanations support auditing and reduce bias in the investigator's judgement. The second takeaway is that the choice between augmenting and replacing the human investigator changes who is accountable, which affects how this project frames its final recommendation. The third takeaway is that this publication supports the decision to favour interpretable

	models and feature importance analysis so that the results can be defended to a compliance team.
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Dataset for Part B

The following publicly available dataset which was used in Publication 1 will be used for the modelling in Part B. It was chosen as a proxy for the compliance-alert data that STATIOEquities cannot provide, because it shares the same core structure. Both are large, high imbalanced, transaction-level binary classification problems in which a rare positive class must be separated from a large majority of legitimate transactions

Name	Credit Card Fraud Detection dataset (ULB Machine Learning Group and Worldline)
Harvard reference	Dal Pozzolo, A., Caelen, O., Johnson, R.A. and Bontempi, G. (2015) 'Calibrating probability with undersampling for unbalanced classification', 2015 IEEE Symposium Series on Computational Intelligence, pp. 159–166. The dataset is distributed on Kaggle by the Machine Learning Group of ULB (2018).
Link	https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
Number of instances	The dataset contains 284,807 transactions in a single file, and has no predefined training and test split. Of these transactions, 492 are fraudulent, which is 0.172 percent of the total, and the remaining 284,315 are legitimate. This is a severe class imbalance. A stratified training and test split will be created during Part B
Number of feature columns	The dataset has 30 original feature columns. These are Time, Amount and V1 to V28. The V1 TO V28 columns are anonymised principal components that were produced by a PCA transformation applied for confidentiality. A further column names Class holds the target label and is not counted as a feature
Number of classes	There are 2 classes, so this is a binary classification problem. Class is equal to 1 for a fraudulent transaction and 0 for a legitimate transaction.
Target range	This criteria does not apply because the task is classification rather than regression.
Origin	The data consist of real anonymised transactions made by European cardholders over two days in September 2013. It was collected and analysed during a research collaboration between Worldline and Machine Learning Group of ULB (Universite Libre de Bruxelles). Because of confidentiality, 28 of the 30 features were transformed into principal components using PCA before the data was released, and only Time and Amount remain in their original form. The data is real and was not synthetically generated.

